Chapter 1

Reproducibility

This appendix says where the code and the records are, which versions of which repositories produced them, how each study is rerun, and where the unmerged parent pull requests stand. Paths are relative to the repository root. No record contains a machine path, and none of the commands needs one. The statements about pins are those of the paper's data-availability section and of `docs/status.md` at commit `3d9eb3a`, the head of `main` when this chapter was written.

1.1 Repositories and pins

The work is spread over four repositories, and the dependence between them is part of what a reader needs to reproduce a number.

Dark-SPECTRAX (`uwplasma/Dark-SPECTRAX`) holds the dark-photon extension, all the study scripts and records, `docs/results.md` (generated from the records by `docs/make_results.py`), `docs/status.md`, and the paper in `paper/` whose figures, numbers and tables are generated by `paper/make_figures.py`. No release or tag has been made. The records and scripts of the paper are those of commit `eb36319`; the relativistic grid records carry `a5aab9f`, the relativistic Hermite records `0873d10` and the nonrelativistic studies `a3ef7a6`.

SPECTRAX (`uwplasma/SPECTRAX`) is the parent Hermite–Fourier solver. Dark-SPECTRAX does not run against its `main`: it is pinned to an integration branch that merges the open pull requests of the author that the studies need. Three pins appear in the records. The reservoir and coherence study used `3de6b2b`, the strong-drive study `4090fbe`, and the collision studies `35231ab`, all on `integration/dark-baseline`. The relativistic Hermite studies used `6a6606b` on `integration/dark-baseline-2`, which is SPECTRAX main `32f7d81` with the open pull requests merged, then PR 88 (the relativistic option, `inv_c2`), then a commit that makes the pump-frame initial state `F0` work in the relativistic branch (branch `feature/relativistic-pump-frame`, fast-forward merged into the integration branch; it has no pull request of its own because it needs `F0`, which only the integration branch has). The repository records state that the same inputs reproduce bitwise on both integration branches for the checks done. None of these branches is a release.

JAX-in-Cell (pin `2d693cb`) and *Dark-JAX-in-Cell* (pin `d547579`) hold the PIC. The pusher is a relativistic Boris/Vay-family one in JAX-in-Cell with `Solver(relativistic=True)` and shape order 2 or 5; Dark-JAX-in-Cell adds the dark-photon drive and the paper-replay example. The grid solver of Chapter **??** uses NumPy and SciPy only.

The environment for the production runs was a CPU virtual environment with JAX 0.6.2 and Python 3.10 on the shared compute host for the PIC and the grid, and the laptop (Python 3.13) for the first relativistic Hermite runs. The GPU resolution study used JAX 0.11.2 with CUDA 12 in double precision. The GPU/CPU comparison for one short case differs by 9.7×10^{-9}.

1.2 Rerunning each study

Every study follows the same pattern: a script declares its gates and its run list and is committed before any run (`*_gates.py`); a driver or batch script runs the cases in chunks that can be resumed from a checkpoint; an analysis script produces a summary record in JSON; and `docs/make_results.py` turns the summaries into `docs/results.md`. The heavy runs take the lock `/tmp/spectrax-heavy.lock` (a directory created with `mkdir`) so that at most one runs at a time on a shared machine. The commands below are the entry points, not a transcript; the arguments are in the scripts' docstrings.

The nonrelativistic strong-drive and truncation studies are `studies/nxconv_*.py` (Nx convergence against the spectral grid), `studies/highk_*.py` (the truncation instability, the closure, the Floquet and eigenvalue checks), `studies/nxconv/batch11.sh` (the wide-box grid reference), `studies/gpures_gates.py` with `gpures_batch.sh` (the GPU resolution extension), and `studies/rclosure_gates.py` with `rclosure_analysis.py` (the closure-strength study). The three rejected alternative bases are in `studies/dai`, `studies/sw` and `studies/antisym`.

The finite reservoir and coherence study is `studies/r1_common.py` (gates and linear theory), `r1_reservoir.py` and `r1_coherence.py`; the strong-drive windows and the seed back-reaction are `studies/r1nl_gates.py`, `r1nl_run.py` run through `r1nl_batch.sh` (200-unit chunks), `r1nl_analysis.py` and the follow-up `r1nl_followup.py`. Collisions are `studies/r2_gates.py`, `r2_batch.py WORKERS [filter]` (a worker pool, each case chunked and resumable), `r2_analysis.py` and `r2_followup.py`, and the LBO-G comparison `studies/h08_lbog.py`. Inhomogeneous conversion is `studies/r3_gates.py`, `r3_ref.py` (the independent cold reference), `r3_kin.py` with `r3_kin_batch.sh`, `r3_analysis.py`, and the follow-ups declared in `r3_followup_gates.py` with `r3_followup_batch.sh`.

The relativistic grid is `studies/relgrid.py` with `relgrid_gates.py`, `relgrid_ref.py` (the exact homogeneous ODE, the cold fluid and the Jüttner dispersion), `relgrid_driver.sh` (gate C2 and the queued parts), `relgrid_batch.sh` and `relgrid_batch2.sh`, `relgrid_office.sh` (the runner used on the shared host) and `relgrid_analysis.py`. The relativistic Hermite model is `studies/relherm/` (gates and the 1D2V checks), `studies/relhhs_gates.py`, `relhhs_run.py` through `relhhs_batch.sh NX NN C` with `SPECTRAX_SRC` pointing to a checkout at `6a6606b`, and `relhhs_analysis.py`; its design note is `docs/design/relativistic-hermite.md`. The k-range study is `studies/hkr_gates.py`, `hkr_grid.py`, `hkr_hermite.sh`, `hkr_par.sh` and `hkr_analysis.py`.

The PIC studies are `studies/relpic/gates.py`, `relpic_run.py`, `batch.sh` and `batch_parallel.sh` (the first twelve runs and the Newtonian controls), then `conv_gates.py`, `conv_followup_gates.py`, `batch_conv.sh TAG CORES LOGDIR` (one tag of the ladder per call, with the environment variables `DJIC`, `JIC` and `PY` naming the Dark-JAX-in-Cell checkout, the JAX-in-Cell checkout and the interpreter) and `conv_analysis.py`. The digitization of the HHS figure is in `studies/relpic/digitization/`. The single-particle pusher checks are in `studies/relpic/single_particle.json`. The paper figures and numbers regenerate with `python paper/make_figures.py` from the repository root.

The tests (`tests/`) run with `pytest -n 2`, 91 tests locally at the last relativistic commit, and the continuous-integration workflow runs them on two Python versions. Heavy PIC and grid runs are not part of the tests.

1.3 Record index

Table 1.1 maps the chapters of this part to their records.

Table 1.1: Records behind the chapters (paths relative to the repository root).

Topic	Records
Windows, truncation instability	`studies/nxconv/summary.json`, `studies/highk/eig.json`, `studies/gpures/summary.json`, `studies/rclosure/stagef.json`, `studies/r1nl/stage1_summary.json`
Seed back-reaction, reservoir, coherence	`studies/r1nl/stage2_summary.json`, `followup_summary.json`, `studies/r1/reservoir_summary.json`, `coherence2_summary.json`
Collisions, LBO-EM and LBO-G	`studies/r2/summary.json`, `studies/h08/summary.json`, `studies/h08_lbog/`
Conversion at a ramp	`studies/r3/` (reference and kinetic records, follow-ups F and G)
Relativistic grid	`studies/relgrid/summary.json`, `homog.json`, `gates_*.json`, `runs/`
Relativistic Hermite	`studies/relherm/gates_*.json`, `studies/relhhs/summary.json`, `gates_*.json`, `runs/`
k-range grid	`studies/hkr/summary_grid.json`, `studies/hkr/validation*.json`
PIC and HHS digitization	`studies/relpic/conv_summary.json`, `records/`, `digitization/hhs_fig2.json`, `single_particle.json`
Status and failed checks	`docs/status.md`, `docs/results.md`, Table II of the paper

1.4 Parent pull requests

The pull requests of SPECTRAX that the studies depend on were all opened by the author and none, at the time of writing, is merged except as noted. The list is from `docs/upstream-review.md`, updated at the resync of 2026-10-25, when main was at `32f7d81`. Merged into main by the owner since the first review: 46, 48, 12, 13, 50 (at `b668d87`) and 86 (Python 3.10 support, which supersedes the `requires-python >=3.11` set by 48). Open and carried on the integration branch: 9, 55, 44, 18, 51, 21, 34, 62, 63, 64, 65 (output, state, configuration and input features), the adaptive-stepping base and 56 to 61 (remapping and pump frame), 66 (field-scaled closure, used by every strong-drive study), 72 (its magnetic option), 68 to 70, 71 (the Dougherty collision operator, LBO-EM), 73 (LBO-G, $\beta = 0$) and 88 (relativistic Hermite). PR 21 still conflicts with PR 18, which both implement a midpoint integrator. The PR 88 code has no syntax that requires Python 3.11. A full test run of the merged integration branch gave 247 passed and one expected failure under Python 3.12; under Python 3.10 with JAX 0.6.2 one test segfaults inside XLA compilation, and it does so on a clean `main` too. The records of the studies stay valid across the repins because the repins only add features that are off by default; the check of bitwise equality for a seeded pump-frame run before and after the repin of 2026-10-16 is in `docs/status.md`. Nothing was merged into SPECTRAX main by the studies, and no pull request was closed.

Bibliography